Customer Shopping Behaviour Analysis

1. Project Overview

This project explores customer shopping behaviour using transactional data from 3,900 purchases across multiple product categories. The objective is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
  - Customer demographics: Age, Gender, Location, Subscription Status
  - Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Colour
  - Shopping behaviour: Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- **Missing Data:** 37 values in the Review Rating column

3. Exploratory Data Analysis (Python)

- **Data Preparation:** Loaded dataset with pandas, checked structure with .info() and summary statistics with .describe().
- **Missing Data Handling:** Imputed missing review ratings using median values per product category.
- **Column Standardization:** Converted to snake_case for readability.
- **Feature Engineering:**
  - Created age_group column by binning ages.
  - Derived purchase_frequency_days from purchase history.
- **Data Consistency:** Verified redundancy between discount_applied and promo_code_used; dropped the latter.
- **Database Integration:** Cleaned dataset stored in PostgreSQL for structured analysis.

SQL Analysis:

We should use SQL in this project because it allows structured and efficient analysis of the large transactional dataset, turning raw purchase records into clear business insights. but SQL is useful for queries like revenue by gender and age group, customer segmentation, and discount dependency. This ensured data integrity, scalability, and reproducibility of results. The insights gained through SQL directly supported recommendations such as loyalty programs, discount policy review, and targeted marketing strategies.

Queries:

Revenue by Gender – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

Top 5 Products by Rating – Found products with the highest average review ratings.

item_purchased	Average_Product_Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

DashBoard in Power BI:

Dashboard Highlights:

- **Subscription Status:** Pie chart showing 27% subscribers vs. 73% non-subscribers.
- **Revenue by Category:** Clothing generates the highest revenue (~100K), followed by Accessories, Footwear, and Outerwear.
- **Sales by Category:** Clothing leads in sales (~1,500 units), with Accessories, Footwear, and Outerwear trailing.
- **Revenue by Age Group:** Young Adults contribute the highest revenue (~60K), followed by Middle-aged, Adults, and Seniors.
- **Sales by Age Group:** Young Adults also lead in sales (~1,000 units), confirming their strong purchasing activity.

Business Recommendations:

- **Increase Subscriptions**: Offer exclusive perks like faster shipping or loyalty points to convert more of the 73% non-subscribers into subscribers.
- **Strengthen Loyalty Programs**: Reward repeat buyers, especially Young Adults who drive the highest revenue and sales, to retain them as loyal customers.
- **Optimize Discounts**: Limit discounts on products already in high demand (like Clothing) and use them strategically for categories with lower sales.
- **Highlight Best-Sellers**: Promote top-performing categories such as Clothing and Accessories in campaigns to maximize revenue impact.
- **Targeted Marketing**: Focus on Young Adults and Middle-aged groups, as they contribute the most revenue, and tailor offers based on shipping preferences using dashboard slicers.

Acknowledgment:

This end-to-end data analytics project, including datasets and guidance, was learned and developed from **Amlan Mohanty's Channel**. It provided the complete workflow — Python for data cleaning, SQL for analysis, Power BI for visualization, and GitHub for portfolio publishing.

References link: https://www.youtube.com/watch?v=5PrZvPeUw60&list=PLAx-M6Di0SisFJ1rv5M_FRHUlGA5rtUf_&index=4